Final Business Recommendations

Bank Marketing Campaign & Customer Response Analytics ·

1. Summary of the evidence

Of 11,162 customers contacted, 47.4% subscribed to the term deposit. Across the three sprints the clearest drivers were: a **previous successful campaign** (91.3% subscribe again), **no loan burden combined with being under 30 or over 60** (about 77%), **housing-loan holders** (36.6% vs 57.0%), **repeated calls** (29.3% at 6+ calls vs 53.4% at one), and an **unknown contact method** (23.8%). Marketing-channel ROI differences were *not* statistically established. All recommendations below come from observational data, so those marked **pilot** should be run as controlled tests before roll-out.

2. Recommendations by area

Customer targeting

R1. Call previous successes first and treat previously contacted customers as warm leads.

- Evidence: previous success converts at 91.3% (95% CI 89.5–92.9, 1,063 customers; chi-square V = 0.30, p = 1.4e-214). Previously contacted customers respond at 67.1% vs 40.7% for new ones.
- Track: response rate of the priority list vs the 47.4% baseline.

R2. Build the main acquisition list around debt-free customers aged 60+ or under 30.

- Evidence: 1,420 customers (12.7% of the base) respond at 77.5% (CI 75.3–79.6). On held-out data, the best age × debt × history segments reached 74.5% and captured 39% of subscribers using 25% of contacts, versus 57.0% and 35% for the best jobs.
- Track: response rate and subscribers per 1,000 calls for this list.

Customer segmentation and product promotion

R3. Use a different offer, or a lower call priority, for customers with a housing loan, especially blue-collar workers. (Pilot)

- Evidence: 36.6% vs 57.0% without a housing loan, a gap of 20.4 points (CI 18.6–22.2). Blue-collar customers with a housing loan respond at 32.7% (CI 30.2–35.3) and are 1,306 customers (11.7% of the base). They still convert about one in three, so they should be pitched differently, not dropped.

- Test: A/B test a loan-sensitive offer against the standard script; compare response and cost per subscriber.

R4. Adopt the engineered features that earned their place for segmentation: age_group, balance_tier, contact_frequency_band, previous_success, previously_contacted, contact_channel_known and debt_status; drop quarter and day_period.

- Evidence: feature evaluation in the Statistical Analysis Report (better segments; quarter and day_period added no value).

Contact strategy

R5. Pilot a cap of about three call attempts, or a channel switch after the third. (Pilot)

- Evidence: 18.0% of customers (2,013) received 4+ calls and responded at 35.2% (CI 33.2–37.3) vs 50.1% for 1–3 calls; 6+ calls: 29.3%. Under the cost proxy, cost per subscriber rises from 5.61 (one call) to 9.17 (6+ calls). This is an association, since customers who refuse tend to be re-called, so only a pilot can show what capping does.
- Track: response rate after the cap, share of calls beyond the third attempt, cost per subscriber.

R6. Audit how 'contact method' is recorded before changing channel strategy.

- Evidence: unknown-contact customers respond at 23.8% (CI 22.2–25.6) vs 54.2% by cellular (V = 0.25), and 92% of those records fall in May and June, so this may be a record-keeping or lead-source problem rather than a channel effect. They also cost about twice as much per subscriber (12.74 vs about 5.75, proxy).
- Track: share of records with unknown contact method.

Campaign planning

R7. Test spreading May volume rather than cutting it blindly. (Pilot)

- Evidence: May holds 25% of all contacts but converts at 32.8%; month has the largest association with response (V = 0.31), but it is confounded with contact mix and campaign intensity, so a controlled test is needed.

R8. Do not move marketing budget between channels on the current evidence.

- Evidence: channel ROI differences are not significant (Kruskal–Wallis p = 0.73) and intervals overlap widely (Email 142–619%). Spend and revenue are almost uncorrelated (+0.06). Collect more campaigns, or run randomised budget tests, before reallocating.

Customer engagement

R9. Track ‘calls ending under 2 minutes’ as a quality KPI and investigate causes (lead quality, script, timing).

- Evidence: calls under 2 minutes convert at 9.3% vs 85.1% for 10+ minutes; duration shows the largest effect found (Cohen’s d = 0.99). Duration is known only after the call and its causal effect is unproven, so it is a diagnostic, not a targeting rule.
- Track: share of calls under 2 minutes by agent, list and month.

Data and ROI foundation

R10. Replace the assumed ROI inputs with real figures and add a campaign ID to every customer contact.

- Evidence: the 0.50 per call-minute and 200 per deposit used here are illustrative. Without a shared key, customer outcomes cannot be tied to campaign cost, so real ROI per customer or channel cannot be computed. Under the proxy, 31.7% of call-time cost goes to customers who decline.

3. Suggested order of work

Phase	Actions
Now (low effort)	R1 priority list, R2 acquisition list, R6 data audit, R9 quality KPI
Pilot (controlled tests)	R3 loan-sensitive offer, R5 call cap, R7 May volume
Foundation	R4 feature adoption, R8 channel evidence, R10 real cost and campaign ID

4. Realistic expectations

Ranking customers with pre-call information improves the odds by about 1.6 times in the top 30% of the list, but contacting only the top half would still miss about 31% of subscribers. Use the ranking to set the **order and the pitch**, not to exclude customers outright.

5. Caveats

- Findings are associations from one historical campaign; pilots are needed to confirm effects.
- Some segments are small and carry wide confidence intervals.
- ROI figures are assumption-based; the 88-campaign marketing table is small and not linked to customers.
- Call duration cannot be used to choose whom to call.